Not-So-Cleansing Recessions

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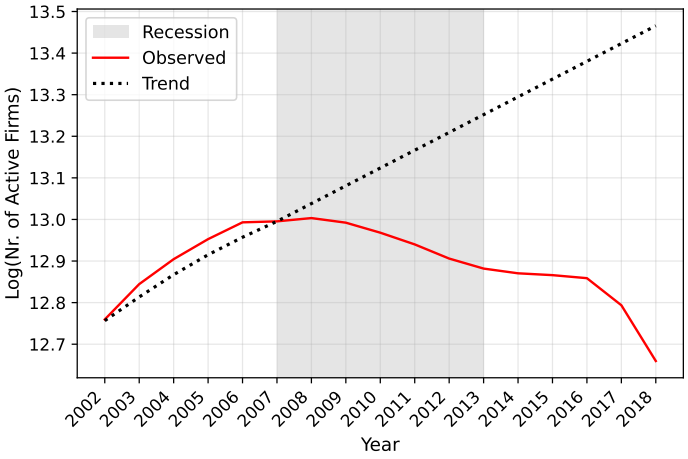
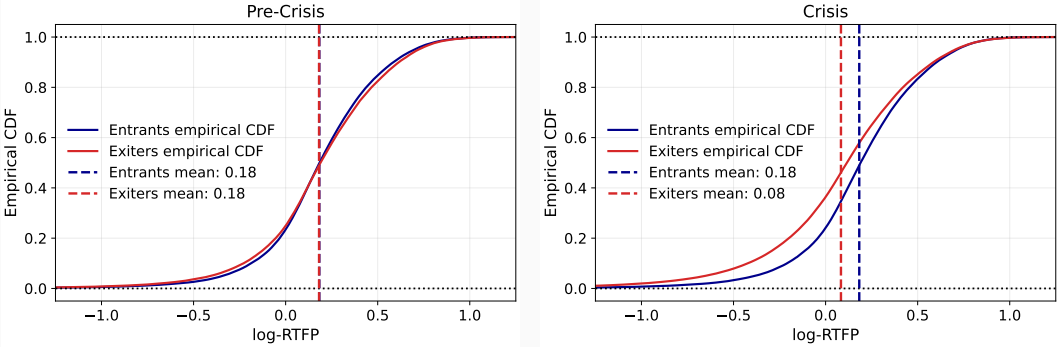
Motivation

Recessions induce large and persistent firm exit



Motivation

While churn occurs all the time, in recessions there is stronger selection



Motivation

This is really at the bottom of the recurrent troubles of capitalist society. ***They are but temporary. They are the means to reconstruct each time the economic system on a more efficient plan.*** *But they inflict losses while they last, drive firms into the bankruptcy court, throw people out of employment,* ***before the ground is clear and the way paved for new achievement of the kind which has created modern civilization and made the greatness of this country.***

— Joseph A. Schumpeter on the Great Depression, 1934

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$\Rightarrow$ *Cleansing effects of recessions*

- Speed up the replacement process/ improve the average efficiency
  Caballero & Hammour (1994)

This Paper

- Theory

  $\Rightarrow$ Recessions induce a trade-off:
  average productivity $\uparrow$ and the number of varieties available $\downarrow$

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- Empirics
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    → substantial costs even in steady-state
- Quantitative model
  - ⇒ Optimal fiscal policy is countercyclical and can almost *halve* the cost of business cycles

Literature

- *Cleansing/Not cleansing recessions*

  Stiglitz (1993), Caballero and Hammour (1994), Barlevy (2002), Ouyang (2009)

  ⇒ New mechanism through which recessions affect welfare: the availability of varieties

- *Macroeconomic effects of firm dynamics*

  Hopenhayn (1993), Bilbiie, Ghironi and Melitz (2012), Clementi and Palazzo (2016), Carvalho and Grassi (2019), Ferrari and Queirós (2024)

  ⇒ Temporary shocks have quasi-permanent outcomes (hysteresis)

- *Love-of-variety estimation* Ardelean (2006), Baqaee, Burstein, Duprez and Farhi (2025)

  ⇒ new approach to estimate LoV based on foreign demand shifts

Model

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- **Environment:**
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- **Exit:** Firm dies if profit turns negative: $\pi(z, m) < 0$.

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- single aggregate price $A(\mathbf{p})$ is a sufficient statistic of the competitive state of the economy
- Nests CES (more later)
- Implicitly defined through an adding up constraint (AUC)

$$1 = \int s\left(\frac{p(z)}{A(\boldsymbol{p})}\right) m(z),$$

where $s(x)$ is the market-share function, which is strictly decreasing in *normalized prices* $x = p/A(\boldsymbol{p})$

Final Good

Suppose some exogenous expenditure $\mathcal{I}$ on the industry, then final good output

$$Y = \upsilon(M) \cdot \mathcal{I} \cdot A(\boldsymbol{p})^{-1} \cdot \exp\left[\int s\left(\frac{p(z)}{A(\boldsymbol{p})}\right)\Phi\left(\frac{p(z)}{A(\boldsymbol{p})}\right) m(z)\right]$$

where

- $\Phi(x)$ is the additional productivity gain associated with a firm whose price is x
- $\upsilon(M)$ is a measure of external effects of varieties M

Equilibrium

Definition

Equilibrium is given by a triplet $\underline{z}$, E, A such that

$$\pi\left(\underline{z}^{-1}/A\right) = 0 \tag{ZPC}$$

$$f^e \geq \int_{\underline{z}}^{\infty} \pi\left(z^{-1}/A\right)\mu^E(z) \tag{FEC}$$

and after entry

$$m_t(z) = \mathbb{1}_{[\underline{z}_t,\infty)}(z) \cdot m_{t-1}(z) + \mathbb{1}_{[\underline{z}_t,\infty)}(z) \cdot E_t \cdot \mu^E(z), \quad (z \geq 0), \tag{LOM}$$

Characterization

Lemma (Invariance with Entry)

Let $\boldsymbol{\phi} = (f^c, f^e, \mathcal{I})$ be a parameter vector and m_{t-1} and m'_{t-1} be two distributions of incumbents. Suppose, the economy given $(\boldsymbol{\phi}, m_{t-1})$ and the economy given $(\boldsymbol{\phi}, m'_{t-1})$ both attain an equilibrium with entry ($E_t > 0$). Then, the aggregate price statistic and cutoff coincide in both economies: $A = A'$ and $\underline{z} = \underline{z}'$.

$\Rightarrow$ economies with the same parameters **but different distributions** have the same marginal firm and state of competitiveness **provided entry is positive**

Business Cycles

Business Cycle - Definition

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- $\tau = 2$: fixed costs unexpectedly increase to $f_h^c > f_l^c$
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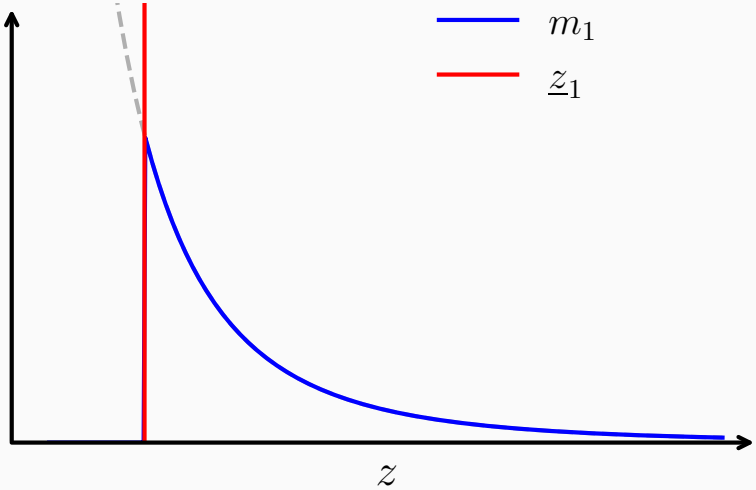
Goal: compare phase 1 to phase 3.

Business Cycle in Pictures Details



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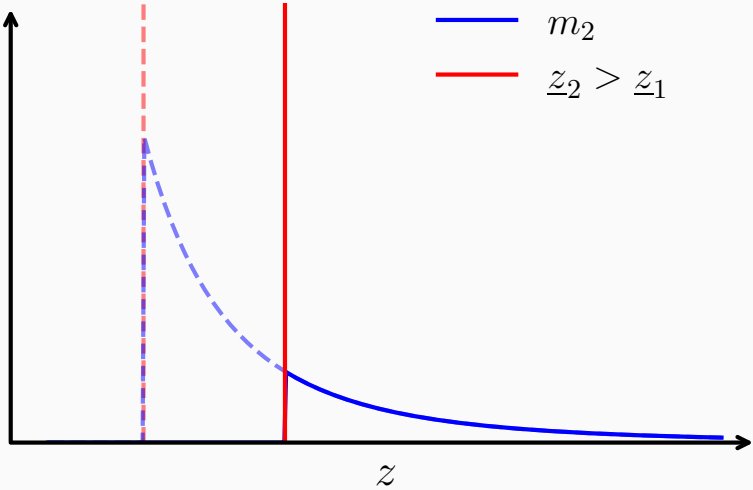

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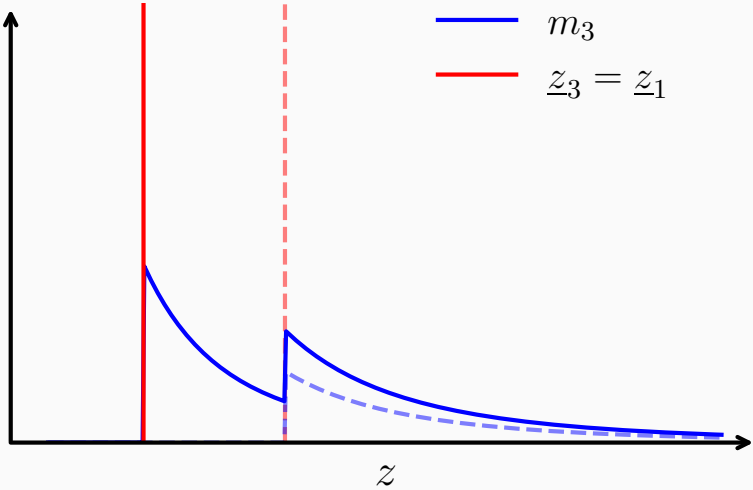

Business Cycle - Results

Theorem (Fundamental Trade-Off of Recessions)

Let $\zeta_E(z) \equiv \frac{\mu^E(z)\,\mathbb{I}\{z \geq \underline{z}_1\}}{\mathbb{P}_{\mu^E}(z \geq \underline{z}_1)}$ *and* $\zeta_X(z) \equiv \frac{m_1(z)\,\mathbb{I}\{\underline{z}_1 \leq z \leq \underline{z}_2\}}{\int_{\underline{z}_1}^{\underline{z}_2} m_1(z)}$ *denote the successful entrant and exiter probability densities, respectively, and let* $\mu_1(z) \equiv \frac{m_1(z)}{M_1}$ *and* $\mu_3(z) \equiv \frac{m_3(z)}{M_3}$ *denote the pre- and post-recession productivity probability densities, respectively.*

- *If* ζ_E *strictly first-order stochastically dominates* ζ_X*:* $\zeta_E \succ_1 \zeta_X$*, then the post-recession mass of varieties falls and the productivity distribution mass shifts right:*

$$\zeta_E \succ_1 \zeta_X \implies \Delta M < 0 \text{ and } \mu_3 \succ_1 \mu_1.$$

- *Conversely, if* ζ_X *strictly first-order stochastically dominates* ζ_E*:* $\zeta_X \succ_1 \zeta_E$*, then the post-recession mass of varieties rises, and the productivity distribution mass shifts left:*

$$\zeta_X \succ_1 \zeta_E \implies \Delta M > 0 \text{ and } \mu_1 \succ_1 \mu_3.$$

Business Cycle - Results

Theorem (Cleansing Effects of Cycles)

Let m_1 be the equilibrium distribution of an equilibrium with entry. Then

$$\Delta \log Y = \underbrace{\Delta \log v(M)}_{\Delta^{ext}} - \underbrace{\Delta \log A(p)}_{\substack{\text{change in} \\ \text{aggregate price}}} + \underbrace{\overbrace{\Theta_1(\underline{z}_1, \underline{z}_2)}^{\text{rev. share weight}} \left[\overbrace{\mathbb{E}_{\theta_1^E}\left[(\Phi \circ X)\left(\frac{1}{zA_1}\right)\right]}^{\text{avg. entrant contribution}} - \overbrace{\mathbb{E}_{\theta_1}\left[(\Phi \circ X)\left(\frac{1}{zA_1}\right) \,\middle|\, \underline{z}_2 \geq z \geq \underline{z}_1\right]}^{\text{avg. exiter contribution}} \right]}_{\Delta^{int} = \text{change in average surplus per variety}}, \tag{1}$$

where the first term, Δ^{ext}, is the effect of the cycle through the variety externality, the second term, $\Delta \log A = 0$ by Lemma 1.

Business Cycle - Remarks

- Last term $\Delta^{int} = 0 \iff$ CES
- If there are no incumbents before phase 1, then the sign of Δ^{int} depends on whether the aggregator has Marshall's Second Law of Demand:

  $MSLD \implies \Delta^{int} > 0$ when recessions are cleansing

Final Good - Special Case

Special case of HSA aggregator above

$$Y = M^{q-\frac{1}{\sigma-1}} \left[\int y(z)^{\frac{\sigma-1}{\sigma}} m(z) dz \right]^{\frac{\sigma}{\sigma-1}}$$

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CES (Dixit and Stiglitz, 1977)

CES if $q = \frac{1}{\sigma-1}$

Business Cycle - Isoelastic Aggregator

Proposition (Cleansing Effects of Cycles)

The change in output after the crisis is given by

$$\Delta \log Y = \left(q - \frac{1}{\sigma - 1}\right) \Delta \log M + \frac{1}{\sigma - 1} \Delta \log \left(\int z^{\sigma - 1} m(z) \right).$$

The recession reduces the number of firms but leaves aggregate productivity unchanged:

$$\Delta \log M < 0 \text{ and } \Delta \log \left(\int z^{\sigma - 1} m(z) \right) = 0.$$

Hence

$$\Delta \log Y \gtreqless 0 \Leftrightarrow q \lesseqgtr q^{CES}.$$

Extras

In the paper:

- General equilibrium Go
- Alternative sources of business cycle Go
- Elastic factor supply Go
- Heterogeneous externality contributions Go
- Forward-looking exit decisions Go
- Analytical aggregate dynamics (transitions) Go
- Idiosyncratic productivity shocks Go
- Multiproduct firms Go
- Alternative units for fixed cost payments Go

Summary

So far

- Temporary recessions permanently change the economy
- Long-run effects on output and welfare depend fundamentally on whether

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Not today: Policy choice also depends on the same condition $q \lesseqqgtr \frac{1}{\sigma-1}$

- Quantifying q and σ — Empirics
- Quantitative model — Quantitative Model

Empirics

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  - Spanish Administrative Firm-level Data (SABI): 2000-2018.

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- **Data:**
  - World Input-Output Database (Timmer et al., 2015): 56 industry, 44 countries, 15 years I-O data
  - Spanish Administrative Firm-level Data (SABI): 2000-2018.
- **Challenge:** Number of varieties, M, is **not observable** in data

Identification of $\frac{1}{\sigma-1} = q^{CES}$

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- Estimate based on Spanish Admin Data is identical

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$$\frac{\partial \log Y}{\partial \log \mathcal{I}} = q \frac{\partial \log M}{\partial \log \mathcal{I}} + 1 + \frac{\partial \log \bar{z}}{\partial \log \mathcal{I}},$$

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- incumbents were not selected through past fixed cost cycles, or
- cleansing effects of past recessions have subsided by random death-shocks, or
- learning.

Identification of q

Proposition (Identification of q)

*Let Δ be the first difference operator. Under **A1**,*

$$\frac{\partial_+ \log Y}{\partial_+ \log X} = q \frac{\partial_+ \log M}{\partial_+ \log X} + 1 + \frac{\partial_+ \log \bar{z}}{\partial_+ \log X}, \quad X \in \{\mathcal{I}, L^p\},$$

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Hence, regress

$$\Delta \log Y_{i,t} = \beta \Delta \log X_{i,t} + \epsilon_{i,t}, \quad X_{i,t} \in \{\mathcal{I}_{i,t}, L^p_{i,t}\},$$

to obtain

$$\widehat{\beta} = 1 + q.$$

Instrument

- Need plausibly exogenous, positive shifts in expenditure for each (r, i, t)
- Ferrari (2024) shift-share instrument
- Identify effect of interest through changes in foreign demand on other products
    - Assign foreign aggregate demand changes to producers based on *exposure shares*

Instrument Details

Threats to Identification

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  $\Rightarrow$ robust to any homogeneous production function with arbitrary homogeneity

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- No **A1**

  $\Rightarrow$ can always recover the sign of $q - 1/(\sigma - 1)$ consistently

Results

Results: q^{CES}

Estimated median q^{CES}: **0.23**

	WIOD			Spanish administrative data (SABI)		
Percentiles	**Profit Rate**	$\boldsymbol{\sigma}$	$\mathbf{q^{CES}}$	**Profit Rate**	$\boldsymbol{\sigma}$	$\mathbf{q^{CES}}$
1	0.0%	.	.	-12.1%	.	.
5	10.0%	10.0	0.11	3.6%	27.6	0.04
10	12.2%	8.2	0.14	6.8%	14.6	0.07
25	13.0%	7.7	0.15	11.7%	8.5	0.13
50	17.2%	5.8	0.21	18.6%	5.4	0.23
75	23.8%	4.2	0.31	28.4%	3.5	0.40
90	31.3%	3.2	0.46	40.8%	2.5	0.69
95	34.4%	2.9	0.52	49.7%	2.0	0.99
99	69.9%	1.4	2.32	65.3%	1.5	1.88

Results: q

- Estimate:

$$\Delta \log Y^i_{c,t} = \beta \Delta \log X^i_{c,t} + \delta_t + \gamma_i + \epsilon^i_{ct}$$

- $X = \{\mathcal{I}, L^p\}$
- Instrumenting $\Delta \log X^i_{c,t}$ with $\widehat{\eta}^i_{ct}$
- Only on positive shocks
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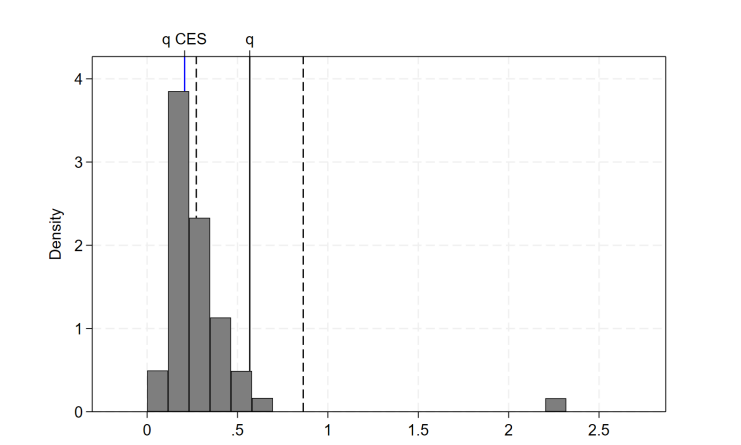
	(D)		(E)		(F)
	(1)	(2)	(3)	(4)	(5)
	$\Delta \log \mathcal{I}^i_{c,t}$	$\Delta \log Y^i_{c,t}$	$\Delta \log L^{p,i}_{c,t}$	$\Delta \log Y^i_{c,t}$	$\Delta \log Y^i_{c,t}$
$\eta^{+,i}_{c,t}$	0.0890***		0.101***		
	(0.0209)		(0.0192)		
$\eta^{+,i}_{c,t-1}$	0.0967***		0.0646***		
	(0.0225)		(0.0191)		
$\Delta \log \mathcal{I}^i_{c,t}$		1.460***			**1.535*** **
		(0.174)			(0.141)
$\Delta \log L^{p,i}_{c,t}$				1.648***	**1.535*** **
				(0.236)	(0.141)
$P(q \leq q^{CES})$		.074		.031	.010
Year FE	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes
N	10067	10067	10083	10083	10067
F-Stat	25	70	25	49	
R^2	0.0323	0.0864	0.0426	0.0881	
F-Stat eq.(1)					1300
F-Stat eq.(2)					127
R^2 eq.(1)					0.0866
R^2 eq.(2)					0.0875

Robust Standard Errors (HC1).

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Results: q vs q^{CES}

3SLS estimates: q and the distribution of q^{CES} across industries:



Summary

Overall $q - q^{CES} > 0$

Next

- Calibrate a model economy to characterize the order of magnitude of long-term effects fixed cost cycles may have.

Conclusions Policy

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- Schumpeter is half right:

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  $\Rightarrow$ Recessions always induce *cleansing* in terms of average productivity

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- Planner chooses to *undo* shift the costs of the recessions to avoid firm exit.

Policy

Two Normative Results

Natural question: what should a (myopic) planner do?

Planner Problem

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Proposition

- *The steady state of the economy is constrained efficient if and only if* $q = \frac{1}{\sigma - 1}$.
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Corollary

A subsidy/tax θ *to the fixed cost is a sufficient instrument to restore efficiency.*

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$\Rightarrow$ Subsidy: $\theta^{ss} > 0 \iff q > 1/(\sigma-1)$.

$\Rightarrow$ Then, also $\theta^c > 0$: subsidize even more in recessions.

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Proposition (Optimal Policy over the Cycle)

The optimal fixed cost taxes/subsidies are counter-cyclical:

$|\theta_1^c| \leq |\theta_2^c|$ *and* $|\theta_3^c| \leq |\theta_2^c|$.

Normative Analysis

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$\Rightarrow$ Later, a quantitative version of the model:

- forward-looking firms: exit based on perfect foresight NPV
- random probability of firm exit δ
- planner accounts for entry costs along the transition

$\Rightarrow$ First, need to quantify q and σ

Quantitative Model

Calibration

- Welfare: $\mathcal{U}(C) = \sum_t \beta^{*t} \log C_t$.

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- Firms discount at rate β where $\beta^* = \beta(1-\delta)$: $\beta = .864$ from Gormsen and Huber (2025).

Calibration

- Fixed cost follows

$$f_t^c = f_{ss}^c + \epsilon\alpha^t$$

- Recessions: a sudden increase in the fixed cost ϵ such that, absent cycle policy response, same exit rate as in Spanish data for the period 2008-2011 in deviation from trend: -20.26%.
- shock reverts following an AR(1) with persistence parameter α

  $\Rightarrow$ generate a half-life of 1 year.

Table

Normative Analysis - Policy

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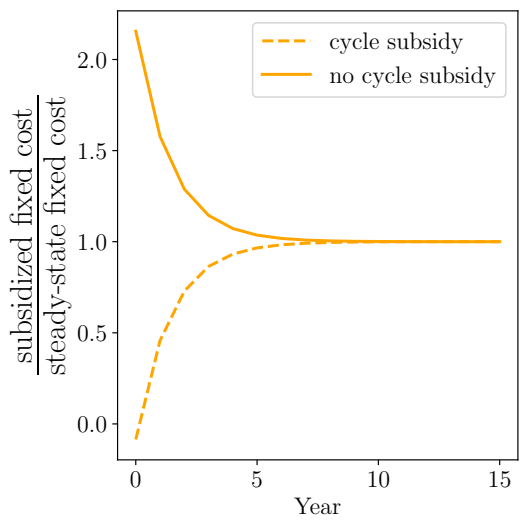
- $\theta^{cyc} > 0 \Rightarrow \widetilde{f}_{ct} < (1 - \theta^{ss}) f_{ct}$: gvt absorbs part of the increase
- $\theta^{cyc} = 1 \Rightarrow \widetilde{f}_{ct} = (1 - \theta^{ss}) \bar{f}_c$: gvt absorbs all of the increase
- $\theta^{cyc} > 1 \Rightarrow \widetilde{f}_{ct} < (1 - \theta^{ss}) \bar{f}_c$: gvt makes effective fixed costs temporarily smaller

Inactive steady state policy



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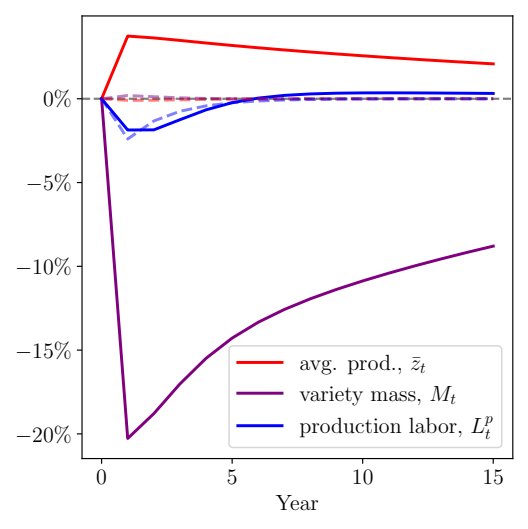

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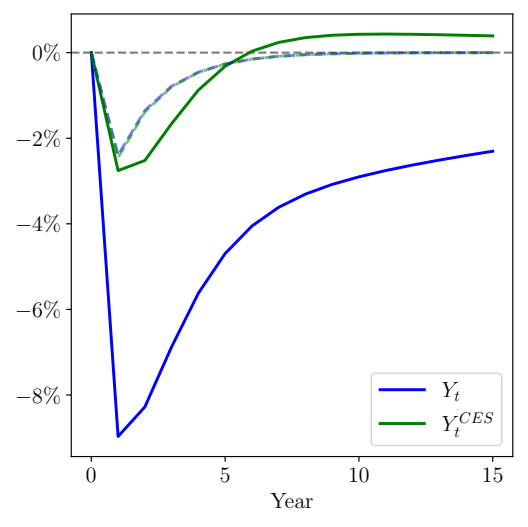

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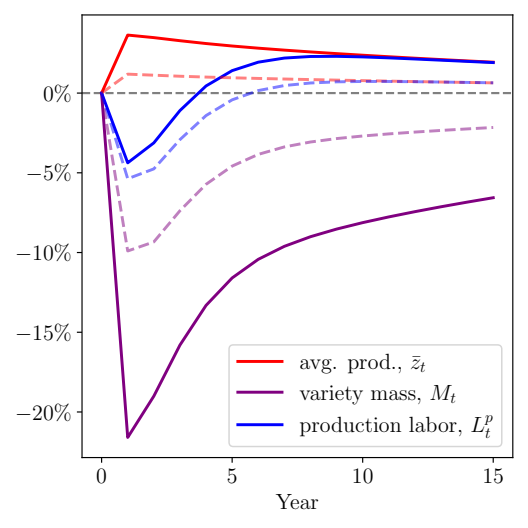

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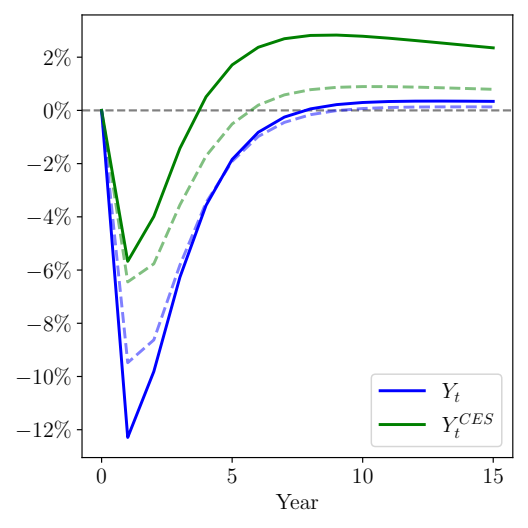

Policy - Welfare

What are the welfare costs of recessions and the benefits of policy?

Welfare Costs of Recessions

Component	CEV relative to Steady State	θ_{ss}	θ_{cyc}
Total Output	−4.57%	inactive	inactive
Varieties	−6.98%		
Production Labor	−0.38%		
Avg. Productivity	+2.78%		
Total Output	−0.62%	inactive	active
Varieties	+0.02%		
Production Labor	−0.61%		
Avg. Productivity	−0.03%		
Total Output	−3.34%	active	inactive
Varieties	−6.10%		
Production Labor	+0.16%		
Avg. Productivity	+2.61%		
Total Output	−3.20%	active	active
Varieties	−2.49%		
Production Labor	−1.56%		
Avg. Productivity	+0.85%		

Policy - Summary

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  $\Rightarrow$ Countercyclical fiscal policy

  Similar to what many governments did during Covid-19

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- Planner optimally *undoes* firm exit by increasing subsidies

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- Optimal subsidy makes the recession significantly shorter as firms do not need to slowly re-enter

Conclusions

References

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Appendix

HSA Back

Proposition (Homothetic Aggregators)

Let $Y(\boldsymbol{y})$ be a homothetic production function from the HSA, HIIA, or HDIA families by Matsuyama (2025), acting on a bundle of inputs $\boldsymbol{y}$. Let $M = |\boldsymbol{y}|$ be the number of varieties in the input bundle. Suppose that

1. *One can separate Y into a variety externality and an aggregator with constant, zero LoV:*

$$Y(\boldsymbol{y}) = h(M)g(\boldsymbol{y}) \quad \textit{with} \quad \frac{d\ln g}{d\ln M} = 0$$

2. *The variety externality has constant LoV:*

$$\frac{d\ln h}{d\ln M} = q = const.$$

Then Y is the aggregator in eq. 1, with
$g(\boldsymbol{y}) = Y^{CES}(\boldsymbol{y})\, M^{-\frac{1}{\sigma-1}}$ *and* $h(M) = M^q$.

Dynamic Interpretation

Remark (Forward-Looking Firms)

Suppose that:

1. *Firms know that the time-t path of* $\{f_t^c\}_t$ *is weakly decreasing.*
2. *Firms calculate the present discounted value of their profit stream.*
3. *Firms receive one-shot offers on whether to enter. If they take the offer, they pay the fixed costs of entry* f^e*, draw their productivity* z*, and can then delay production until they become profitable.*

For large t*, the measure of firms in the economy is the same as in the case of myopic firms.*

Path Dependency

Remark (Path Dependence)

The stationary steady-state equilibrium is path-dependent.

- We considered economies that, in phases 1 and 3, feature identical parameters. Nonetheless, they are characterized by different equilibrium allocations.
- This property is fully driven by the presence of incumbents.

Path Dependency of Recessions

Remark (Path Dependence of Recessions)

Let a q-economy experience two cycles of the same intensity f_c^h. Then:
a) the post-crisis distribution, output, and welfare are different across the two cycles;
b) the recessions generate different degrees of cleansing, captured by $\underline{z}_2^1 \leq \underline{z}_2^2$, where $i \in \{1, 2\}$ denote the cycle.

- Having experienced the first recession, the q-economy starts the second crisis with a different incumbent population.
- During downturns, the composition of incumbents determines the cutoff for a given increase in the fixed cost f_c^h.

Lemma 1 - Intuition

ZPC is an optimal exit condition:

$$\pi(\underline{z}, m) = 0 \Leftrightarrow f^c = \widetilde{\pi}(\underline{z}, m)$$

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In CES: $\frac{\widetilde{\pi}(a,m)}{\widetilde{\pi}(\underline{z},m)} = \left(\frac{a}{\underline{z}}\right)^{\sigma-1}$: independent of the incumbents

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- As f^c is high
  - $M \downarrow$
  - $\bar{z}_t \uparrow$

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Cleansing Effects

Extensions

Business Cycle - General Eqm

- in GE:
  - Budget Constraint: $\mathcal{I} = R = (w)\bar{L} + \Pi$
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- Then $\Delta \log Y > 0$ in CES, and

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$$\Delta \log Y = \left(q - \frac{1}{\sigma - 1}\right) \Delta \log M + \Delta \log L^p + \Delta \log \mathcal{Z}$$

$-$ $\quad$ $+$ in GE $\quad$ $+$

- Then $\Delta \log Y > 0$ in CES, and
- $\exists\,!\, q^\star > \frac{1}{\sigma - 1} : \Delta \log Y = 0$.

Back

Alternative Sources of Business Cycles

- so far, the *most cleansing* recessions possible: movements in f_c which directly control selection

Back

Alternative Sources of Business Cycles

- so far, the *most cleansing* recessions possible: movements in f_c which directly control selection
- Alternatives:
  - Aggregate TFP
  - movements in f_e
    (need to change the model to have a positive steady-state entry)

Back

Alternative Sources of Business Cycles -

Back

Proposition (Aggregate TFP Cycles)

Recessions driven by temporary decreases in aggr. TFP have no long-run effects.

Alternative Sources of Business Cycles -

Back

Proposition (Aggregate TFP Cycles)

Recessions driven by temporary decreases in aggr. TFP have no long-run effects.

Proposition (Entry Cost Cycles)

Consider a temporary increase in the entry cost. Along this transition, the following holds

a) *The number of varieties converges to* M^{ss} *from below.*

b) *The average productivity of active firms converges to* $\bar{z}^{ss}$ *from below.*

c) *The temporary increase in the entry cost always generates welfare losses, independently of love-of-variety.*

Elastic Labor Supply

Proposition (Elastic Factor Supply)

Consider an economy with elastic primary factor supply (EF). Then, relative to an economy with fixed factor supply (FF), long-run output effects of a recession are dampened: $|\Delta \log Y^{EF}| < |\Delta \log Y^{FF}|$.

Back

Heterogeneous Externality

- suppose that each firm draws an externality contribution ξ upon entry, jointly with z from $\mu^E(\xi, z)$ with $\mathbb{E}_{\mu^E(\xi)} = 1$.
- Define the variety effect aggregator $\mathcal{M} \equiv \int\int \xi m(z, \xi)\xi = \mathbb{E}_{m_\xi}[\xi] \times M$
- and the final good production function

$$Y = \mathcal{M}^q \times \left[\int y(z)^{(\sigma-1)/\sigma} \mu_z(z)\right]^{\frac{\sigma}{\sigma-1}}.$$

Remark (Independence)

If ξ and z are independent in entrant and incumbent distribution, then the model is equivalent to our baseline model in firm-level choices and aggregates.

Back

Heterogeneous Externality

Proposition (Business Cycles with Heterogeneous Externalities)

Consider the extension with heterogeneous contributions to the externality. Say that ξ and z are positively (negatively) correlated if $\mathbb{E}_{\mu^E}[\xi \mid z \geq a]$ is strictly increasing (decreasing) in a. Let Y^{hom} be output in our baseline model with homogeneous externalities. Then:

1. *The cycle induces negative selection in ξ and $\Delta Y < \Delta Y^{hom}$ if ξ and z are negatively correlated.*
2. *The cycle induces positive selection in ξ and $\Delta Y > \Delta Y^{hom}$ if ξ and z are positively correlated.*
3. *The cycle induces no selection in ξ and the heterogeneity is irrelevant to output $\Delta Y = \Delta Y^{hom}$, if ξ and z are uncorrelated.*

Back

Forward-looking Exit

- Suppose firms anticipate that fixed costs revert but not future entry, along the transition.
- Then the exit rule is

$$NPV(\{f^c_{t+k}\}_k) \gtreqless \widetilde{\pi}(z, m_t)/(1-\beta).$$

- The recession is less deep as fewer firms exit on impact but qualitatively the same.

Back

Aggregate Dynamics

- Suppose the productivity distribution upon entry is Pareto: $\mu^E(z) = \beta z_{min}^{\beta} z^{-\beta-1}$ with $\beta > \sigma - 1$
- normalize initial fixed costs $f_{-1}^c = 1$
- Suppose fixed costs follow

$$f_t^c = \phi^{\max\{0, 1 - t \cdot \frac{1}{T^*}\}}, \quad (t = 0, 1, 2, ...)$$

- Then the transition can be characterized analytically (see paper) and

Back

Aggregate Dynamics

Back

Proposition (Continuous Time Reversion)

In the limit as $T^* \to \infty$, the mass of active firms converges from above to

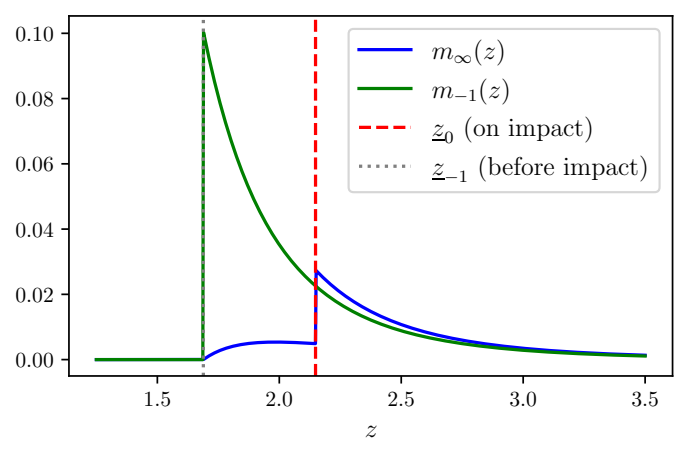
$$M_{T^*} \downarrow M_\infty = \frac{\mathcal{I}}{\sigma}\frac{\beta-(\sigma-1)}{\beta}\left((1-\phi^{-1})\frac{\beta-(\sigma-1)}{\beta}+\phi^{-1}\right).$$

Moreover, the firm distribution converges to

$$m_{T^*}(z) \to m_\infty(z) = \underbrace{\mu^E(z)\overbrace{\frac{\sigma-1}{\beta}\frac{\mathcal{I}}{\sigma}\frac{1}{f^e}}^{=E_{-1}}}_{=m_{-1}(z)} \times \begin{cases} (\beta-(\sigma-1))\left[\ln z - \ln \underline{z}_{-1}\right] & z \in [\underline{z}_{-1}, \underline{z}_0), \\ \left[1+\frac{\beta-(\sigma-1)}{\beta}\ln\phi\right] & z \in [\underline{z}_0, \infty). \end{cases}$$

The right tail of $m_{T^*}(z)$, $z \geq \underline{z}_0$ converges to $m_\infty(z)$ from below. The limit density $m_\infty(z)$ is continuous, except at $z = \underline{z}_0$, where it jumps up.

Aggregate Dynamics



Back

Idiosyncratic Productivity Shocks

- Suppose firms face idiosyncratic productivity shocks such that their productivity is $\widetilde{z} = z + \varepsilon$ with $\varepsilon \sim \psi$ with $\mathbb{E}[\varepsilon] = 0$
- then the eqm is given by

$$
\begin{aligned}
& m_t(z) = m_{t-1}(z)\mathbb{I}_{\{z \geq \underline{z}_t\}} + E_{\varepsilon,t}\mu^E(z)\mathbb{I}_{\{z \geq \underline{z}_t\}}, \\
& E_{\varepsilon,t}(\mathbb{E}_{\mu^E}[\max\{\mathbb{E}_\psi[\pi(z, m_t)], 0\}] - f^e) \leq 0, \\
& E_{\varepsilon,t} \geq 0, \\
& \mathbb{E}_\psi \pi(\underline{z} + \varepsilon, m_t) = 0,
\end{aligned}
$$

Back

Idiosyncratic Productivity Shocks

Proposition (Equilibrium with idiosyncratic risk)

The following holds

1. *As in the baseline model, the cutoff, $\underline{z}$ is history-independent in a state of entry.*
2. *An incumbent with permanent productivity component z is part of the steady-state if and only if the probability that temporary productivity fluctuations cannot make them exit at the peak of the crisis: $\mathbb{P}_\psi(z + \varepsilon < \underline{z}) = 0$. The pre-crisis and long-run allocations differ if and only if*

$$\int \mathbb{P}_\psi(z + \varepsilon < \underline{z})\mu^E(z) > 0.$$

A sufficient condition for both these to be true is that there exists a lower bound $\underline{\varepsilon}$ high enough.

Back

Multiproduct Firms

- Suppose firms hold multiple products $N(z)$ and firm-level output is given by the aggregator

$$y(z) = \left(\int_0^{N(z)} y_j(z)^{\frac{\eta-1}{\eta}} j \right)^{\frac{\eta}{\eta-1}}.$$

  with elasticity of substitution within the firm $\eta > \sigma > 1$

- Then the HSA aggregator with the properties above is

$$Y = M^{q_F - \frac{1}{\sigma-1}} \widetilde{N}^{q_V - \frac{1}{\eta-1}} \left[\int y(z)^{\frac{\sigma}{\sigma-1}} m(z) \right]^{\frac{\sigma-1}{\sigma}}.$$

  where $\widetilde{N} \equiv N/M$

Back

Multiproduct Firms

Back

Proposition (Fixed-cost Cycles with Multi-Product Firms)

In the presence of multi-product firms, the following holds true.

1. *For recessions driven by temporary increases in f^c, where $q_F^{CES} := \frac{1}{\sigma-1}$ and $q_V^{CES} := \frac{1}{\eta-1}$:*

   1.1 *an analogue of Proposition 1 holds with*

$$M_3 < M_1,\ L_3^p = L_1^p,\ N_3 = N_1,\ \mathcal{Z}_3 = \mathcal{Z}_1$$

   and

$$\frac{Y_3}{Y_1} \gtreqless 1 \Leftrightarrow q_F - q_F^{CES} \lesseqgtr q_V - q_V^{CES};$$

   1.2 *an analogue of Proposition 2 holds with*

$$M_3 < M_1,\ L_3^p > L_1^p,\ N_3 > N_1,\ \mathcal{Z}_3 > \mathcal{Z}_1$$

Multiproduct Firms

Back

Proposition (Fixed-cost Cycles with Multi-Product Firms)

and there exists a unique $q_F^\star$ *with*

$$q_F^\star - q_F^{CES} > q_V - q_V^{CES},$$

such that

$$\frac{Y_3}{Y_1} \gtreqless 1 \iff q_F \lesseqgtr q_F^\star;$$

1.3 *an analogue of Proposition 5 holds with*

$$\exists!\,(q_\circ, q^\circ) \;\; \textit{with} \;\; q_\circ - q_F^{CES} > q_V - q_V^{CES}$$

such that long-run effects depend on f_h^c *intensity.*

2. *Recessions driven by temporary increases in* f^p *have no long-run effects.*

Output Fixed Costs

Back

- Suppose fixed and entry costs are in a new bundle made by

$$B(Y, L) \;=\; \kappa(\alpha)\, Y^{\alpha} L^{\,1-\alpha}, \qquad \kappa(\alpha) = \alpha^{\alpha}(1-\alpha)^{1-\alpha}, \quad \alpha \in [0,1]$$

with $\sigma - 1 > \alpha$

Proposition (Cleansing Effects of Cycles)

The change in output after the crisis, when the fixed costs are produced by a Cobb-Douglas bundle of output (share α) and labor (share $1-\alpha$), is given by

$$\Delta \log Y(q, \alpha) = \frac{\sigma - \alpha}{\sigma - 1}\, \Delta \log \left(\int z^{\sigma-1} m(z) \right) + (1-\alpha)\,(\, q - q^{\mathrm{CES}}\,)\, \Delta \log M.$$

Output Fixed Costs

Back

Proposition (Cleansing Effects of Cycles)

For each $\alpha \in [0,1]$ there exists a value of love-of-variety $q^\star(\alpha)$ such that the crisis leaves output unchanged: $\Delta \log Y(q^\star(\alpha), \alpha) = 0$; with $q^\star(\alpha) \geq q^{\mathrm{CES}}$, with equality for $\alpha = 1$. Furthermore, output is decreasing in q around $q^\star(\alpha)$: $\frac{\partial}{\partial q} \Delta \log Y(q, \alpha)\Big|_{q=q^\star(\alpha)} < 0$. Which implies

$$q \gtrless q^\star(\alpha) \Rightarrow \Delta \log Y(q) \lessgtr 0 \quad \textit{for } q \textit{ in a neighborhood of } q^\star(\alpha)\textit{, to first order.}$$

Policy

Planner Problem

Suppose the planner solves

$$\max_{E_t^{SP}, \underline{z}_t^{SP}} \quad Y_t^{SP} = (M_t^{SP})^{q-\frac{1}{\sigma-1}} L^{p,SP} \left[\int z^{\sigma-1} m_t^{SP}(z) \right]^{\frac{1}{\sigma-1}} \qquad \text{s.t.}$$

$$\bar{L} \geq L_t^{p,SP} + f^e E_t^{SP} + f^c M_t^{SP},$$

$$m_t^{SP}(z) = (m_{t-1}(z) + E_t^{SP} \mu^E(z)) \mathbb{I}_{\{z \geq \underline{z}_t^{SP}\}}(z),$$

$$\underline{z}_t^{SP}, E_t^{SP} \geq 0.$$

Back

Depth of Recessions

Business Cycle Depth

So far: Fix a cycle f_l^c, f_h^c and see for which q cycle is welfare improving.

Business Cycle Depth

So far: Fix a cycle f_l^c, f_h^c and see for which q cycle is welfare improving.

Now: Fix q, q^{CES} at some level and consider now recessions of **different intensities** f_h^c:

$\tau = 1$: fixed costs equal f_l^c

$\tau = 2$: fixed costs unexpectedly increase to $f_h^c = (1+\epsilon)f_l^c, (1+2\epsilon)f_l^c, (1+3\epsilon)f_l^c, ...$

$\tau = 3$: fixed costs revert to f_l^c

(Recall that we are after long-run effects.)

Business Cycle Depth

Given q, q^{CES}...

are deeper crises better in the long run?

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Recessions trade off **variety losses** vs. **cleansing effects**.

Business Cycle Depth

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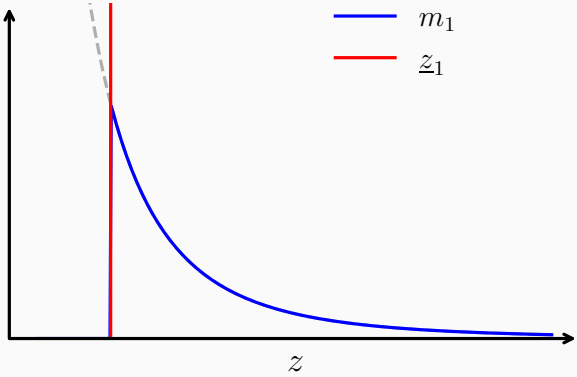
Recessions trade off **variety losses** vs. **cleansing effects**.

A **deeper recession** necessarily generates more exit along the transition but **effect on long-run Y, M etc. is not obvious.**

3 Recessions Example

A Tiny Recession

- ever so slight increase of f^c
- reversion of fixed cost: a sliver of low-productivity firms was killed
- on aggregate, nothing much happened



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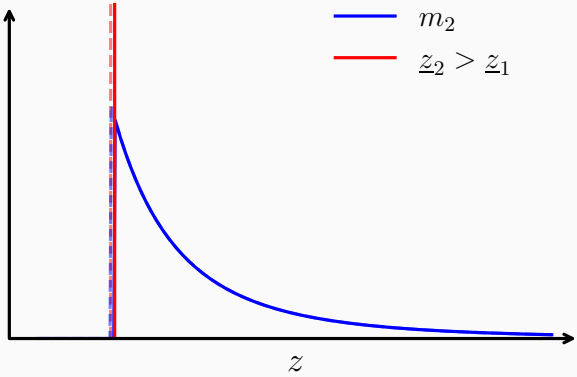

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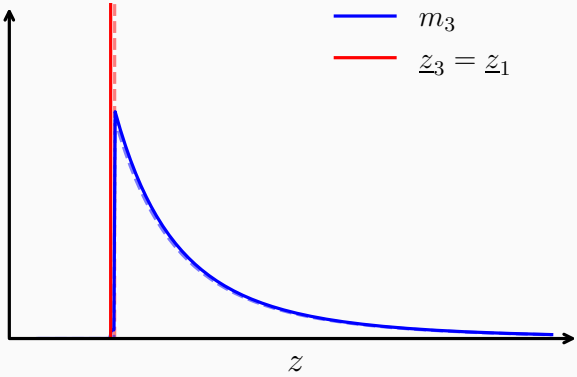
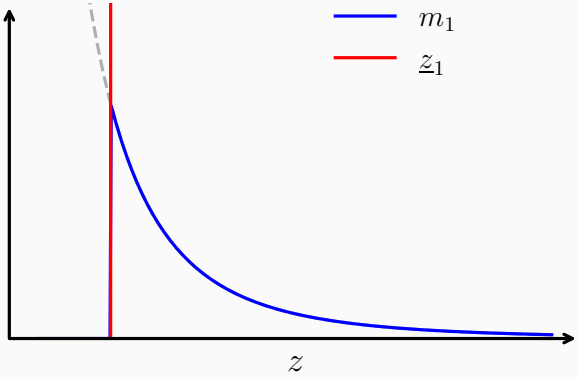


A Huge Recession

- jack up f^c, kill almost all firms
- reversion of fixed cost: new entrants look a lot like firms who were killed!
- on aggregate, again, nothing much happened
- Also **neither cost nor benefit:**



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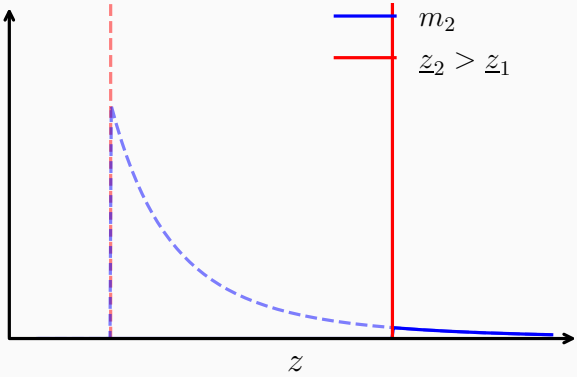

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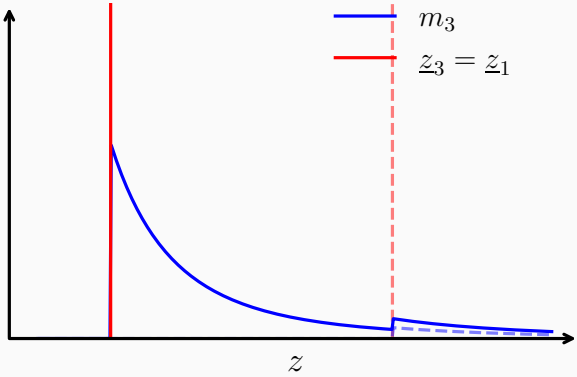

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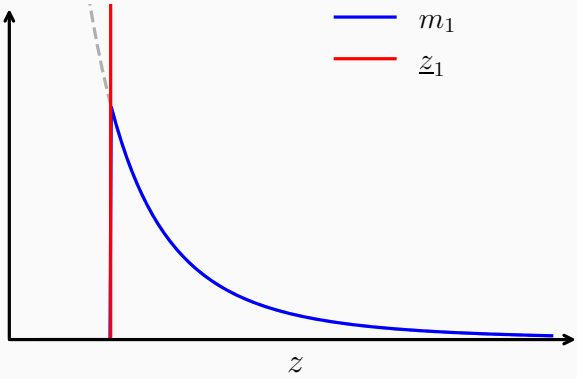
A Moderate Recession

- moderate increase in f^c
- reversion of fixed cost: a **new entrants are better** than firms who were killed!
- Lasting effects. Some signs are clear: $\Delta M^{mod} < 0$, $\Delta L^{d,mod} > 0$, $\Delta \mathcal{Z}^{mod} > 0$, but unsure about ΔY^{mod}.



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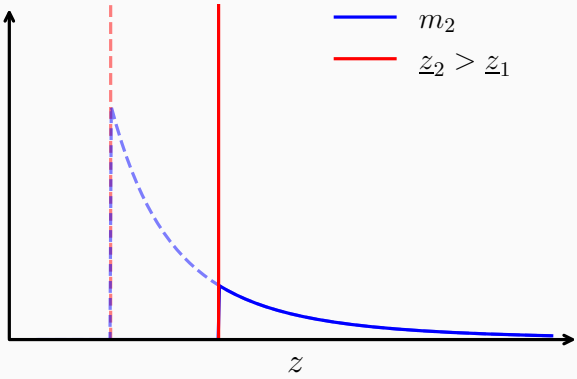

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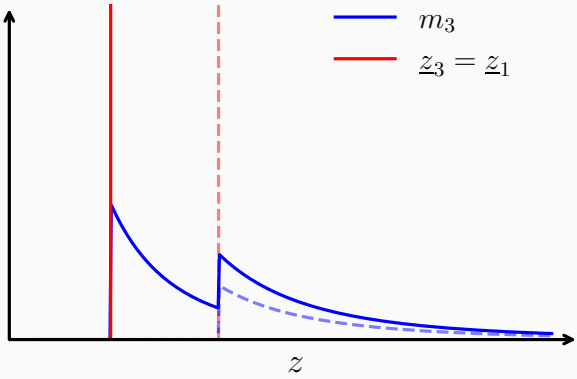

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Formal Output Decomposition

Start again from equilibrium post-recession aggregate output: skip formal decomposition

$$Y_3 = M_3^{q-\frac{1}{\sigma-1}} \underbrace{L_3^p \left(\int z^{\sigma-1} \mu_3(z) dz \right)^{\frac{1}{\sigma-1}}}_{Y_3^{CES}}$$

Then, we have:

$$\frac{\partial \log Y_3}{\partial \log f_h^c} = \underbrace{\frac{\partial \log M_3}{\partial \log f_h^c}}_{(1)} \times \underbrace{\left[(q - q^{CES}) + \frac{\partial \log Y_3^{CES}}{\partial \log M_3} \right]}_{(2)}$$

(Note: $\partial \log Y_3 / \partial \log f_h^c$ is different from $\Delta \log Y$!)

Decomposition of the Elasticity – M_3 to f_h^c

$$\underbrace{\frac{\partial \log M_3}{\partial \log f_h^c}}_{(1)}$$

- Number of firms/varieties necessarily declines after a crisis. Compare entrants (in phase 3) and leavers (in phase 2):

  $\mathbb{E}_0[z|z \geq \underline{z}_1] \geq \mathbb{E}_0[z|\underline{z}_1 \leq z \leq \underline{z}_2]$

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  at high $f_h^c \Rightarrow \underline{z}_2 > \mathbb{E}_0[z|z \geq \underline{z}_1]$

  $\Rightarrow$ marginal exiter more productive than avg. entrant

  $\Rightarrow$ marginal increase in the long-run M

Decomposition of the Elasticity – Y_3^{CES} to M_3

$$\underbrace{\underbrace{(q - q^{CES})}_{(a)} + \frac{\partial \log Y_3^{CES}}{\underbrace{\partial \log M_3}_{(b)}}}_{(2)}$$

- (a) is constant.

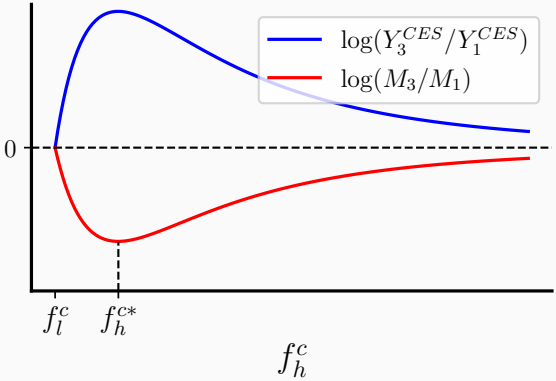
Decomposition of the Elasticity – Y_3^{CES} to M_3

$$\underbrace{\underbrace{(q - q^{CES})}_{(a)} + \frac{\partial \log Y_3^{CES}}{\underbrace{\partial \log M_3}_{(b)}}}_{(2)}$$

- (a) is constant.
- (b) always negative but not monotone
  - when M_3 is large eliminating 1% of firms $\Rightarrow$ large labor savings
  - when M_3 is small $\Rightarrow$ small labor savings

Behavior of CES output and varieties as a function of f_h^c

If $q - q^{CES} = 0$, then ΔM is irrelevant and $\Delta L^p > 0$, and $\Delta \mathcal{Z} > 0$ guarantee that $\Delta Y > 0$, for all f_h^c: skip formal decomposition



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Interaction of Cycle Depth and Love-of-Variety

Are all recessions cleansing?

$$\partial \log Y_3 / \partial \log f_h^c$$
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Depends a lot on q!

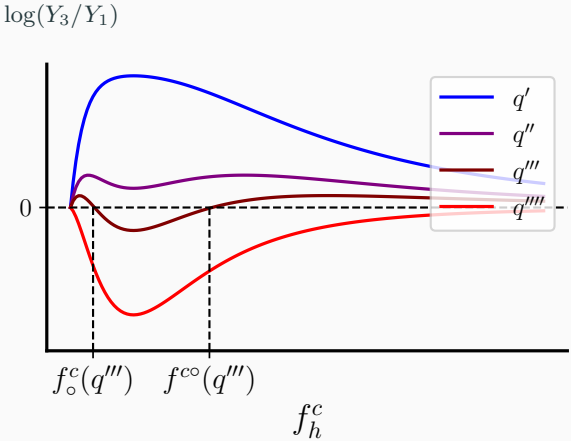
- Large q: (2) is globally **positive**, all recessions are beneficial and some f_h^{c*} is the best-case recession
- Small q: (2) is globally **negative**, all recessions are hurtful and some f_h^{c*} is the worst-case recession
- Some intermediate $q < q^{CES}$: Small effects with unclear sign! There exist both, recessions with beneficial and hurtful long run effects!

Interaction of Cycle Depth and Love-of-Variety

So, are all recessions cleansing?

Taking stock:

$$\partial \log Y_3 / \partial \log f_h^c$$
$$= \underbrace{\frac{\partial \log M_3}{\partial \log f_h^c}}_{(1)} \times \underbrace{\left[(q - q^{CES}) + \frac{\partial \log Y_3^{CES}}{\partial \log M_3} \right]}_{(2)}$$

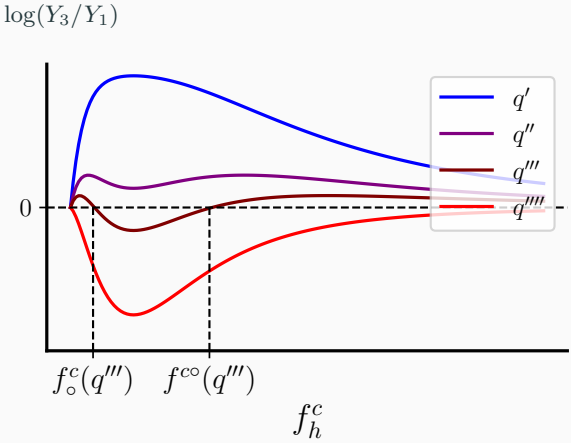


$$q' < q'' < q''' < q''''$$

Interaction of Cycle Depth and Love-of-Variety

Result:

- If q is **low enough**, then all recessions are welfare **improving**



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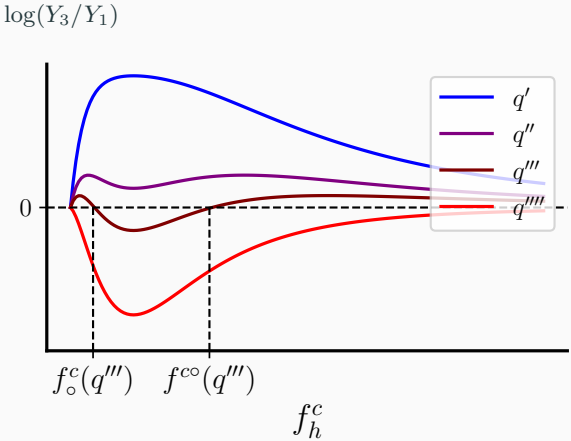
Interaction of Cycle Depth and Love-of-Variety

Result:

- If q is **low enough**, then all recessions are welfare **improving**
- $q = q^{CES}$ falls into this range



$$q' < q'' < q''' < q''''$$

Interaction of Cycle Depth and Love-of-Variety

Result:

- If q is **low enough**, then all recessions are welfare **improving**
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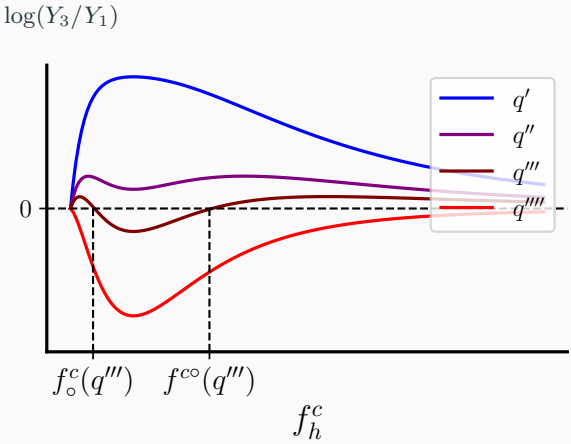

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Interaction of Cycle Depth and Love-of-Variety

Result:

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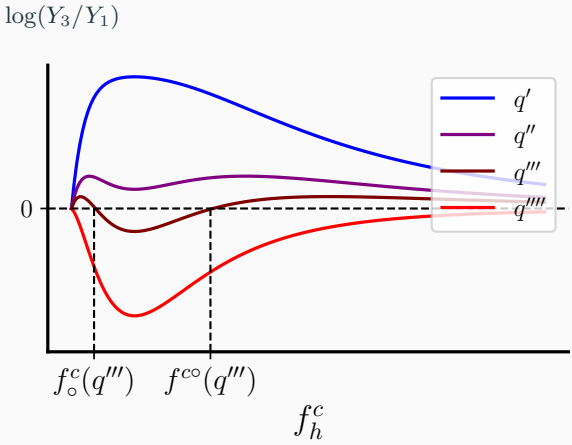
- If q is **high enough**, then all recessions are welfare **decreasing**
- **Intermediate values** of $q \implies$ 'wiggles'

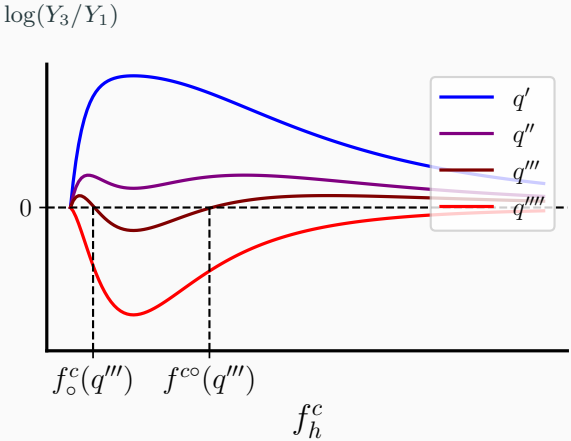


$$q' < q'' < q''' < q''''$$

Interaction of Cycle Depth and Love-of-Variety

Need guidance on q

$\Rightarrow$ **Estimate q and σ**



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Empirics

Instrument

Industry r in country i hit by a (total) demand shock η_{it}^r which is the weighted sum of demand shocks $\eta_{j,t}$ flowing from households in destination countries j

$$\eta_{i,t}^r = \sum_j \underbrace{\xi_{i\to j}^r}_{\text{destination shares}} \times \overbrace{\eta_{j,t}}^{\text{destination shocks}}$$

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- $\xi_{i\to j}^r$share of output of (r,i) that goes (directly or indirectly) into the consumption of households in country j.
- Demand innovations $\eta_{j,t}$ are specific to country j and travel upstream to suppliers

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- Meaning of $\eta_{j,t}$: Average growth rate of sales to final consumers in country j at time t, where the avg is across all the producing industries
- Aggregate the $\widehat{\eta}_{j,t}(i,r)$ using the sales shares into $\widehat{\eta}^{i}_{c,t}$; this is our instrument.

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Calibration Table

	Value	Description	Target/Source
β	0.864	Firm discount rate (p.a.)	Gormsen and Huber (2025)
δ	0.0388	Death rate (p.a.)	pre-GFC exit rate in SABI
β^*	0.894	Social planner discount factor (p.a.)	$\beta/(1-\delta)$
h	1.2	Tail parameter	size dist. in SABI
σ	5.4	Elasticity of substitution	profit rate SABI
q	0.535	Love-of-variety	estimate
L^p	1.0	Labor supply	normalization
f_0^c	1.0	Steady state fixed cost	normalization
α	0.841	Shock decay	1 year half-life
ϵ	see text	Shock magnitude	# firms loss SABI

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